Evolutionary history from genomes to single cells by lossless compression

Sergei L. Kosakovsky Pond[1*], Darren P. Martin[2], Stephanie J. K. Pond[3], Steven Weaver[1], Danielle Callan[1], and Anton Nekrutenko[4]

[1]Institute for Genomics and Evolutionary Medicine,

Department of Biology, Temple University, Philadelphia, PA, USA.

[2]Institute of Infectious Disease and Molecular Medicine,

Department of Integrative Biomedical Sciences,

University of Cape Town, Cape Town, South Africa.

[3]Bioinnovation and Genomic Sciences Division,

Translational Genomics Research Institute, Phoenix, AZ, USA.

[4]Department of Biochemistry and Molecular Biology,

The Pennsylvania State University, University Park, PA, USA.

[*]Corresponding author. Email: spond@temple.edu

Short title: Evolutionary history by lossless compression

One-Sentence Summary: Lossless data compression unifies evolutionary trees, rapid radiations, gene transfer, and convergence across scales of life.

Reconstructing evolutionary history is essential across biology, yet standard pipelines rely on separate models and arbitrary cutoffs to build trees, collapse weak branches, and detect recombination or convergence. Because inheritance copies an ancestor with sparse edits, we recast tree and network inference as

lossless data compression. In Bushwhacker, a branch or lateral link is added only when the shared changes it explains save more information than specifying it costs. Rapid bursts without shared mutations remain unresolved as multiway hubs, horizontal transfers distinguish cleanly from parallel sites, and large phylogenies run orders of magnitude faster than maximum likelihood in far less memory. Across cells, viruses, microbes, and mammals, compression resolves apparent gene conflicts, maps recombinants, disentangles gene conversion from adaptive convergence, and predicts clonal selection.

Evolutionary trees are among the most widely used structures in science: they organize Earth's biodiversity, track unfolding epidemics, trace immune and tumor clones inside individual patients, and anchor comparative genomics across the Tree of Life. Yet even as sequencing has expanded from single genes to millions of genomes, the conceptual foundations used to reconstruct trees have changed remarkably little for nearly half a century. Since distance clustering, parsimony, and probabilistic Markovian likelihood were introduced between the 1960s and early 1980s (*1–3*), major advances have largely focused on scaling tree search and enriching parametric substitution models within those classical paradigms (*4–6*). Genuinely new mathematical perspectives on how evolutionary history can be inferred have been rare.

Here we introduce a different foundation, rooted in information theory: lossless data compression (*7, 8*). Biological inheritance is, at heart, a copying process with sparse edits. Whether an ancestor transmits a genome, a protein, or a somatic cellular profile, related lineages share most of their states because they inherited them from a common source. Recording shared changes once along an ancestral branch is therefore far more economical than recording them separately in every descendant—making an evolutionary graph the natural coordinate system for compressing biological data (Fig. 1). This equivalence is not merely metaphorical: encoding viral and microbial cohorts along an inferred phylogeny physically compresses genomes substantially further than general-purpose algorithms, until pervasive recombination breaks tree-like descent itself.

That insight reframes evolutionary reconstruction around a single question: what is the shortest complete recipe that regenerates every observed genome, protein, or cell from a common ancestor without losing a single character? Starting from one ancestral sequence, our approach builds an evolutionary graph in which each branch transmits only a compact edit list—where changes occurred

and what new states appeared—while every branch or lateral link in the graph adds to the description length. Following Rissanen's Minimum Description Length principle (*7, 8*), every connection must earn its place on one common ledger: a branch or link is added only when the shared edits it eliminates save more information than describing that connection costs. Because description length maps directly onto probability, the information saved by each connection immediately quantifies the odds that it is real—replacing separate model-selection sweeps, pruning cutoffs, and bootstrap resampling with a single self-calibrating score (*9, 10*).

Once every evolutionary claim has to pay its own cost in information, phenomena that normally require incompatible tools fall into place naturally. When cells proliferate, outbreaks super-spread, or species radiate faster than mutations arise, sub-branches have no shared edits to compress; specifying them wastes information, so rapid bursts stay open as multiway hubs (polytomies) rather than inventing unstable binary splits that clash across models and genes (*11–13*). Across uneven rates or billion-year divergences, pinpointing a rare shared change on a quiet ancestral stem saves far more address information than matching chance collisions on fast-evolving branches, defeating long-branch artifacts (*14*) and scaling smoothly to continuous substitution-rate models (*15, 16*). And once vertical descent has compressed everything a tree can explain, residual similarities between distant lineages separate by their genomic footprint: a transferred segment pays just two endpoint coordinates to explain dozens of linked differences—exposing recombination, horizontal gene transfer, and paralog gene conversion (*17, 18*)—whereas adaptive convergence stands out as scattered, high-leverage parallel sites (*19*).

Implemented in our open-source engine Bushwhacker, this formulation unlocks four capabilities beyond existing phylogenetic pipelines. First, it reconstructs Tree-of-Life DNA and protein supermatrices 24- to 54-fold faster than fast maximum-likelihood search (142- to 288-fold faster than approximate maximum likelihood) in 10- to 40-fold less memory, delivering a probabilistically annotated graph—not a bare point-estimate tree—in a single deterministic pass. Second, it naturally preserves rapid evolutionary bursts as multifurcations instead of inventing arbitrary strictly bifurcating splits that flip across substitution models, search perturbations, and loci (*13*). Third, it jointly infers vertical trees and lateral networks, distinguishing transferred segments from scattered convergent sites that otherwise distort clades (*20–22*). Fourth, it generalizes natively beyond nucleotide and protein alignments to discrete single-cell somatic mutations and ordered

copy-number states while compiling genome cohorts into lossless .bwz archives physically smaller than general-purpose compressors. Across eukaryotic radiations, 882 mammalian coding exomes and proteomes (*23*), pandemic viruses, prokaryotic domains (*24, 25*), and single-cell immune and tumor cohorts (*26, 27*), lossless compression resolves apparent gene-tree conflict, separates paralog gene conversion from ecological convergence, maps mosaic recombinants, and predicts in vivo clonal selection directly from sequence or somatic mutation profiles.

Trees and networks by lossless compression

To evaluate how concisely an evolutionary graph explains a set of biological sequences or cellular character profiles, Bushwhacker encodes two parts without loss in natural logarithmic units of information (nats; 1 nat $\approx$ 1.44 bits): the graph itself and the state changes along its edges (Fig. 1A, fig. S1, and table S5) (*28*). Starting from one ancestral sequence at the root, each branch records only how many positions changed, where those changes occurred, and what new nucleotide, amino-acid, or somatic states appeared, weighted by empirical exchange probabilities (*16*). Resolving a multifurcating burst into a new ancestral branch avoids repeating shared changes independently in each descendant, but incurs the address cost of specifying which lineages join the clade and encoding the new branch itself (Fig. 1B). Because every prefix-free code length L defines a probability $P \propto e^{-L}$, comparing nested structures with and without a candidate branch or lateral link on a heuristically optimized graph gives its local log-odds evidence ΔL and conditional two-hypothesis posterior support $P_{\text{local}} = 1/(1 + e^{-\Delta L})$, eliminating the need to evaluate branch support with bootstrap replicates. On a four-taxon burst, zero or one shared mutation costs more to specify than it saves ($\Delta L = -1.8$ and -0.1 nats; $P_{\text{local}} = 14\%$ and 47%), leaving the burst unresolved, whereas two to four shared changes yield positive evidence (+1.2 to +3.9 nats; $P_{\text{local}} = 77\%$ to 98%).

Two further consequences follow from measuring positions and connections in a shared currency (Fig. 1C,D and Supplementary Text). First, specifying an edit's location costs far more on a nearly invariant ("quiet") branch than on a mutation-dense one; consequently, moving a shared change onto an ancestral stem from quiet branches saves substantially more information than grouping coincidental matches on fast-evolving branches (5.6-fold in Fig. 1C), recovering likelihood-like rate

weighting (*2*) from combinatorics alone. When divergence spans hundreds of millions or billions of years and individual sites mutate repeatedly, this discrete code generalizes smoothly within the same two-part code to continuous-time eight-category substitution rate models (*8, 15, 16*), evaluated in compact CPU cache tiles that avoid gigabyte-scale memory tables (*28*). Second, once the vertical tree is established by topological refinement, candidate lateral links between distant lineages are priced by their spatial footprint: a contiguous transferred tract requires only two genomic endpoint coordinates to explain many linked changes, whereas scattered parallel replacements must index each site individually across the variable-site or variable-codon pool, distinguishing block transfers and gene conversion from site-specific adaptive convergence.

When trees hold and when bifurcations fail

Before applying this framework to unresolved biological problems, we asked two foundational questions: can lossless compression match established probabilistic methods when evolution genuinely follows a deep tree, and how does it behave when real histories depart from a single strictly bifurcating tree? Classical deep phylogenies are a stringent test for any combinatorial method because naive change-counting groups fast-evolving lineages by chance collisions on billion-year divergences—the flaw that originally drove phylogenetics toward computationally intensive likelihood models (*2, 14*). Indeed, an unweighted-parsimony version of our engine collapses on deep prokaryotic divisions (Fig. 2A). By contrast, once edits are weighted by evolutionary rate across long-branch benchmarks (*16*) and genome-scale alignments of mammals (*23*), bacteria, and archaea (*25*), lossless compression recovers higher-rank clades alongside fast maximum-likelihood tools (IQ-TREE 2 (*6*) and FastTreeMP (*4*))—matching them on mammalian orders (100%) and bacterial phyla (94.3%) and trailing by a few percentage points on the deepest archaeal splits (tables S2 and S11). Where overall tree distances differ slightly (0.5 to 2.5 percentage points; table S11), the difference reflects how uninformative branches are handled: rather than guessing at short, mutation-poor stems, Bushwhacker leaves them open, trading a small slice of fine-tip recall to halve false splits on mammalian proteins. Because rate-weighted scores are evaluated locally without iterative parameter optimization, thousand-species supermatrices finish in about a minute within laptop memory—one to two orders of magnitude faster than maximum likelihood in a small

fraction of the memory—making whole-proteome tree and network censuses practical.

Evaluating confidence in individual branches ordinarily multiplies that computational cost by requiring hundreds of trees to be re-estimated on resampled alignments. Across more than 44, 000 branches with known ground truth, the information saved by each branch directly gauges its reliability in a single pass (Fig. 2B and table S3). Branches with decisive compression savings ($P_{\text{local}} \geq 99\%$, nearly 90% of retained splits) are genuine 98.3% of the time. On the remaining borderline splits (P_{local} between 50% and 99%), local posteriors are more optimistic than observed accuracy (41% to 80% true) because P_{local} tests each branch against collapsing that single edge on one optimized tree rather than averaging over competing topologies; high P_{local} therefore pinpoints dependable clades directly, whereas modest values flag branches near the resolution limit.

Even well-calibrated tree builders break down, however, when evolutionary history is not a single binary tree—whether through rapid bursts, genetic exchange, parallel adaptation, or noisy single-cell measurements. Across 37, 020 ground-truth simulations spanning nucleotides, proteins, coalescent radiations, lateral networks, and single cells, evaluating every feature by description length handles all three regimes natively (Fig. 2C,D, figs. S2 and S3, and tables S2 to S5). When lineages radiate faster than mutations accumulate, strictly bifurcating likelihood fits chance similarities and invents dozens of false splits per burst—even when statistical polytomy tests are enabled (*13*)—whereas compression leaves mutation-free bursts open as multiway hubs (> 94% split precision) and improves downstream coalescent species-tree inference (*29*). When distant clades exchange DNA tracts or evolve parallel adaptive changes, forcing a single tree pulls convergent lineages together (0% sister-clade retention under five shared convergent changes in maximum likelihood), whereas encoding lateral links on the same ledger keeps the vertical tree intact (99.2% clade retention) and separates transferred tracts from parallel sites (90% to 100% recovery on moderate-to-large events; table S4) with zero false links on tree-like controls. Finally, on single-cell somatic mutation and copy-number profiles degraded by allelic dropout, native multi-state and ordered-step coding improves clonal precision and copy-number recovery over binary likelihood and binarized coding while calling zero false clones on healthy diploid cells.

Mutation-free branches explain gene-tree conflict

Roughly two-thirds of published single-gene trees clash with genome-wide phylogenies, fueling debates over hybridization and lineage sorting (*11, 12*). However, a major share of these clashes arises without biological discordance: merely changing the substitution model or starting search perturbation routinely shifts the tree inferred from the exact same alignment (*13*). Both problems stem from enforcing a strict bifurcation rule onto brief, unmutated lineages. When speciation outpaces mutation in an individual locus, short ancestral stems record zero informative changes; leaving those zero-change stems unresolved ($\Delta L \leq 0$, $P_{\text{local}} \leq 50\%$) states probabilistically that the gene cannot distinguish competing branching orders from a coin flip. On a typical mammalian gene (*CHST13*, 1,176 bp; Fig. 3A), enforced bifurcations create eight conflicting splits that bootstrap support does not cleanly resolve: collapsing near-zero branches removes only one conflict, four errors retain 50–75% ultrafast bootstrap support (UFBoot; pulling bushbabies onto rodents at 63% and sloth onto pika at 62%), and a 95% cutoff discards true Catarrhini (93% UFBoot). By leaving the mutation-free placental hub open, Bushwhacker recovers nine true clades ($P_{\text{local}} = 0.82$–1.0) with zero conflicts in 0.02 s (3.8 s for UFBoot; default 8-category rate heterogeneity retains 10 true clades plus two borderline splits at $P_{\text{local}} = 0.66$–0.69; fig. S7).

Across 45,205 genes in 19 eukaryotic radiations, withholding zero-evidence splits removes estimation noise rather than biological signal (Fig. 3B,C and tables S1 and S6). With no reference tree involved, 95% to 96% of branches that flip across substitution models or independent RAxML/IQ-TREE searches on identical alignments carry $\Delta L \leq 0$ and drop out automatically. Measured against published genome-wide phylogenies, withholding $\Delta L \leq 0$ splits eliminates 61.5% of apparent conflicts (74.9% in early birds (*30, 31*) and 82.5% in early animals (*11*)) while retaining 77.2% of shared clades—nearly invariant to how the reference phylogeny itself was built (*32*)—and compressing translated protein sequences rather than coding DNA where third codon positions are mutationally saturated strips away base-composition biases in deep plant and insect radiations (*33*).

Crucially, because species trees are themselves inferred from genes, forcing mutation-free loci into strictly bifurcating trees creates a circular trap: thousands of uninformative genes cast random coin-flip votes that destabilize contested deep nodes (*12*) (Fig. 3D). Weighting branches by their positive compression support ($\Delta L > 0$) breaks that circularity—resolving lungfish as sister to

tetrapods where unweighted binary gene counts stall in a three-way tie (*34*), uniting turtles with archosaurs > 10-fold, placing ctenophores as the earliest animal lineage (*35, 36*), and favoring Atlantogenata at the placental root (*37*). Passing multifurcating Bushwhacker gene trees directly to ASTRAL-IV (*29*) likewise lowers coalescent species-tree error without bootstrap cutoffs across rapid-speciation "anomaly-zone" simulations—where the most frequent binary gene tree disagrees with the true species tree (table S2)—and 11,164 distinct empirical loci (12,008 alignments) spanning animals, plants, fungi, birds (*38*), and mammals (table S1).

Gene conversion and convergence across mammals

Measuring incomplete lineage sorting, paralog gene conversion, and adaptive convergence together across whole proteomes has been out of reach for likelihood pipelines, which cannot scale to thousands of hundreds-of-species alignments and routinely let unaccounted gene conversion or convergent sites distort single-gene and protein phylogenies. On the outer hair-cell motor gene *Prestin* (*SLC26A5*; 116 mammals), where sonar-linked replacements famously pull echolocating dolphins inside bats on standard protein trees (*19*), Bushwhacker keeps whales inside Cetartiodactyla ($P_{\text{local}} = 1.00$) and bats monophyletic ($P_{\text{local}} = 1.00$; 93.8% species-split precision) while isolating a lateral convergence link (+3.9 nats, $P_{\text{local}} = 98.0\%$) between dolphins and horseshoe bats across four shared motor-domain codons (Fig. 4A and fig. S4). Scaling this framework to 19,952 orthologous coding genes (19,953 proteins) across up to 882 mammals (*23*) shows how strongly a single organismal tree organizes the mammalian proteome (Fig. 4B, fig. S5, and tables S1 and S7): 85.9% of mammalian proteins—rising to 97.9% above 1,500 amino acids—compress more tightly on the mammalian species phylogeny than on an unconstrained gene tree, while the fraction of candidate species-tree branches showing genuine incomplete lineage sorting in proteins (19.8% overall) climbs with protein length toward the 20.8% ancestral polymorphism baseline measured in coding DNA.

Without using synteny or paralog labels, spatial pricing separates two lateral processes across those proteins (Fig. 4C). Contiguous paralog gene conversion is detected in 6.0% of mammalian proteins (median tract 85 residues), acts largely within families (68.9%; 88.3% within orders), and enriches 3.0- to 6.6-fold inside tandemly arrayed clusters—including globins, detoxification

enzymes, olfactory receptors, zinc-finger arrays, and cadherins or keratins—versus 4.2% in other coding loci. Scattered parallel amino-acid changes, by contrast, are inferred in 11.9% of proteins and cross mammalian orders 38.0% of the time (3.3-fold higher cross-order share than paralog gene conversion), identifying candidate sites under convergent selection.

Those cross-order parallel sites reveal that ecological adaptation is sharply focal rather than proteome-wide (Fig. 4D and fig. S5C). Resolving whether sonar drove convergence across thousands of proteins (*20*) or reflects background noise (*21, 39*), echolocating bats, toothed whales, and solenodons show no proteome-wide excess (0.50–0.75× random expectation), and marine (1.31×) and subterranean blind (1.47×) mammals are likewise neutral across the proteome. Even the raw 12.6× excess in armored ant- and termite-eating pangolins, anteaters, and armadillos (four links) traces largely to giant muscle and centrosomal proteins (*TTN*, *ALMS1*; 1.35× after length adjustment) and multicopy receptor or zinc-finger families (*OR10D3*, *ZSCAN16*; table S7). Instead, adaptive convergence concentrates in shared physiological pathways: positive parallel-site compression links appear across 70% (19 of 27 genes; 10 of 27 on the fixed species tree, plus 116-species *Prestin*) of the inner-ear hair-cell mechanotransduction complex—spanning stereocilia tip and ankle links (*USH2A*, *ADGRV1*, *CDH23*, *PCDH15*), the *TRIOBP* rootlet bundler, the *LOXHD1* and *STRC* connectors, and the *TMC1–CIB2* channel complex—much as C_4 photosynthesis concentrates parallel changes in catalytic grass enzymes (*40*).

Outbreak hubs, recombinants, and gene transfer

Pathogen evolution is characterized by processes that are poorly reflected by strictly bifurcating trees: super-spreading transmits identical or single-mutation genomes to dozens of hosts before substitutions accumulate, co-infections facilitate frequent recombination between divergent strains, and selection for immune escape or drug resistance drives parallel adaptive mutations across multitudes of independent lineages (*18, 22*). Whereas forced binary trees invent arbitrary transmission chains inside acute outbreaks that are better represented as polytomies, and likelihood recombination scans cannot scale to pandemic cohorts (*10*), Bushwhacker resolves outbreak hubs, recombinational transfers, and convergent mutations together (Fig. 5A,C and tables S1 and S7). Across 10, 062 SARS-CoV-2 and 5, 498 dengue virus genomes, compression preserves 1, 426 SARS-CoV-2 super-

spreading hubs of up to 335 descendants alongside urban dengue bursts—which nearly double on translated proteins where silent nucleotide changes no longer separate rapid transmissions—while mapping SARS-CoV-2 Spike and Omicron mosaic tracts, natural dengue recombinants (*18*), and HIV-1 recombination breakpoints (*28*).

Encoding genome cohorts into lossless .bwz archives via arithmetic coding also tests directly whether shared descent is the main physical regularity of biological sequences (Fig. 5B and table S9) (*28*). Storing 10,062 SARS-CoV-2 genomes requires 300.9 MB in uncompressed form and only 636 KB as a complete lossless archive (63.2 bytes per viral genome; 1.25× smaller than xz -6 and 11.5× smaller than gzip -9), or 388 KB for the compressed sequence payload alone (up to 2.06× smaller than xz -6 and 18.9× smaller than gzip -9), with consistent gains across influenza, dengue, mitochondrial, plastid, and whole bacterial chromosomes. Equally telling is when that advantage reverses: across controlled recombination titrations ($K = 0 \rightarrow 32$), mosaic bat sarbecoviruses (0.71×) (*41*), and recombinant HIV-1 (0.65×), pervasive whole-genome mosaicism erases any single tree and lets multi-donor string dictionaries overtake single-tree coding, making archive size a direct gauge of tree-like versus mosaic evolution.

In prokaryotes, measuring vertical descent and horizontal transfer in the same information units turns qualitative debates over a microbial "Tree versus Web of Life" (*17*) and the Complexity Hypothesis (*24*) into a quantitative budget (Fig. 5D and table S8). In hospital outbreaks of carbapenem-resistant *Klebsiella pneumoniae* (*42*), the 214-kb core chromosome resolves every branch across nine clinical isolates (6/6 splits at $P_{\text{local}} = 1.00$, with lateral transfer contributing just 0.5% of the chromosomal budget), whereas the 882-bp *blaKPC* plasmid carried by those exact same strains collapses 5 of 6 splits into multifurcating *blaKPC-2* and *blaKPC-3* (C814T) star hubs separated by a single nucleotide as horizontal plasmid spread outpaces mutation. Across 122 marker proteins in 1,672 archaeal species (*25*), horizontal transfers likewise appear exclusively in operational metabolic and regulatory genes (11.8%, 12/102—including cross-phylum highways between symbiotic DPANN archaea and their hosts (*43*)) versus 0% (0/20, two-sided Fisher $p = 0.21$) in core ribosomal and RNA polymerase subunits; and as single-gene microbiome 16S rRNA trees scale from 256 to 32,768 taxa (*44*), supported splits saturate smoothly from 91.4% ($N = 1,024$) to 76.0% ($N = 32,768$), cutting taxonomy-contradicting splits per group by 19% to 24% while running 4.5× faster than FastTreeMP (table S1).

Clonal selection in single cells

Inside germinal centers and tumors, positively selected cells divide so rapidly that a single founder spawns a star-like burst of daughter lineages before new somatic mutations arise (*45*). Whereas specialized lineage tools recover these bursts only when given cell-abundance counts or hypermutation models (*46*), reconstructing 147 mouse germinal centers (10,429 B cells across 29 mice, 9,352 unique genotypes (*26*)) from deduplicated sequences alone shows that compression geometry reads out Darwinian selection directly (Fig. 6A,B and table S1).

Grouping B-cell genotypes solely by how many mutant daughter lineages (k) radiate from each node predicts a 9.5-fold rise in in vivo clonal expansion from $k = 0$ terminal leaves (5.1%) to $k \geq 3$ burst hubs (48.2%; paired Wilcoxon $p = 5.9 \times 10^{-6}$), correlates with tighter yeast-display antigen binding within germinal centers ($p = 1.7 \times 10^{-3}$), and holds within exact mutational-depth strata ($p = 4.1 \times 10^{-6}$) while vanishing in neutral simulated repertoires ($p = 0.70$, table S2; empirical cohorts, table S10). Across mice, mutations arising independently in five or more animals enrich 5.6-fold inside antigen-contacting complementarity-determining regions (64.0% versus 24.1%, $p = 7.5 \times 10^{-5}$) and confer higher binding affinity ($p = 8.1 \times 10^{-5}$; Fig. 6A,B).

Extending the same framework beyond DNA sequences to somatic mutation and copy-number profiles in human cancer—with all cell counts and clinical labels withheld—reveals the same selective signature in patient tumors (Fig. 6A,C,D, fig. S6, and table S10). In a single leukemia patient (AML-38-001, 7,016 sequenced cells; Fig. 6C), reconstructing clonal history in 4.1 ms recovers an early *NPM1* founder that splits into parallel *IDH1* ($P_{\text{local}} = 0.996$, $k = 3$) and *IDH2* ($P_{\text{local}} = 0.997$, $k = 4$) burst hubs, each independently radiating subclonal *FLT3*, *KRAS*, *NRAS*, and *PTPN11* resistance branches, whereas Bayesian Markov chain Monte Carlo (MCMC) sampling (471 s) nests *IDH2* sequentially downstream of *IDH1* (table S10). Across 168 COMPASS myeloid single-cell cohorts (129 patients, 1.00 million cells (*47*)), inferred burst hubs ($k \geq 3$) carry 6.18-fold more leukemic cells in vivo than terminal leaves (232.8 versus 37.7 cells; $p = 1.7 \times 10^{-18}$; exact patient-depth paired $p = 5.9 \times 10^{-12}$) and resolve 90.3% of stepwise copy-neutral loss-of-heterozygosity (CN-LOH) transitions; across the separate 123-patient Morita AML cohort (*27*), branching hubs spawn larger descendant lineages ($p = 3.0 \times 10^{-15}$) while late signaling resistance drivers enrich relative to early epigenetic founders (OR = 1.79, $p = 1.9 \times 10^{-3}$) and recur in

parallel within 44 patients (Fig. 6A,D). On 4,950 single-cell whole-genome copy-number profiles across ovarian and breast tumors (*48*), ordered step coding compresses raw 500-kb genomes 21.9- to 33.5-fold in 22 s on colinear segments, contracting 39%–53% of noisy splits while preserving cohort clade purity (57.9%–95.7% versus 20.8%–59.9% random null; OV2295 ovarian tumor versus ascites 95.7% purity, compartment classification score $F_1 = 0.997$; table S10).

Information as a common evolutionary currency

Viewing evolutionary inference as lossless compression shows that tree topology, branch support, evolutionary rates, gene transfer, and adaptive convergence follow from a single principle: shared descent shortens the description of biological variation. Because branches and lateral links are retained only when they compress the data, unresolved hubs become a quantitative biological signal, marking rapid species radiations, epidemic super-spreading, and Darwinian clonal bursts in immunity and cancer without arbitrary cutoffs. Evaluating vertical clades, transferred tracts, and parallel sites on equal terms separates local paralog gene conversion from cross-order convergence and turns debates over microbial gene transfer into genome-wide budgets, while scaling Tree-of-Life phylogenies to standard laptops and web browsers. Because this principle requires only heritable states copied with sparse changes, it extends naturally beyond sequences and somatic profiles to epigenetic marks, CRISPR lineage barcodes, and morphological or linguistic characters wherever history leaves a compressible trace.

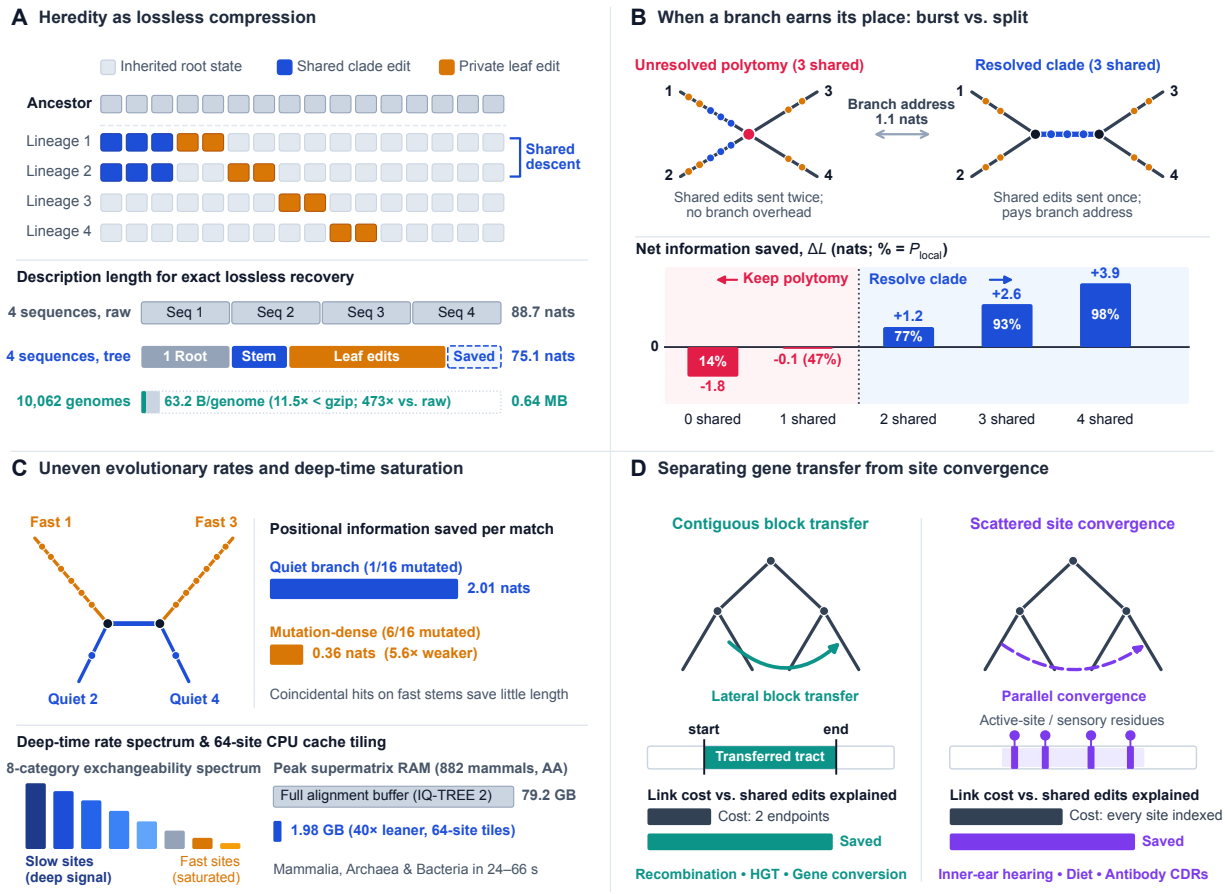


Figure 1: **Evolutionary trees and networks are lossless compression structures.** (**A**) Encoding one ancestral root plus sparse branch edits shortens four 16-site sequences from 88.7 to 75.1 nats (1 nat $\approx$ 1.44 bits), compounding to 63.2 bytes per 29.9-kb genome (11.5× smaller than gzip -9; table S9) across 10,062 SARS-CoV-2 genomes. (**B**) Resolving a burst shares edits across descendants (top) but pays the topological split address (1.1 nats on a quartet). Zero or one shared change loses information (-1.8 and -0.1 nats; $P_{\text{local}} = 14\%$ and 47%) and remains a polytomy, whereas two to four shared changes yield positive net savings (+1.2 to +3.9 nats; $P_{\text{local}} = 77\%$ to 98%). (**C**) Positional coding yields 5.6× greater savings on quiet stems than on mutation-dense branches, while eight-category rate coding in 64-site CPU tiles scales to billion-year supermatrices in under 2.8 GB. (**D**) Contiguous lateral transfers require only two genomic endpoints to explain linked edits, filtering out scattered homoplasy.

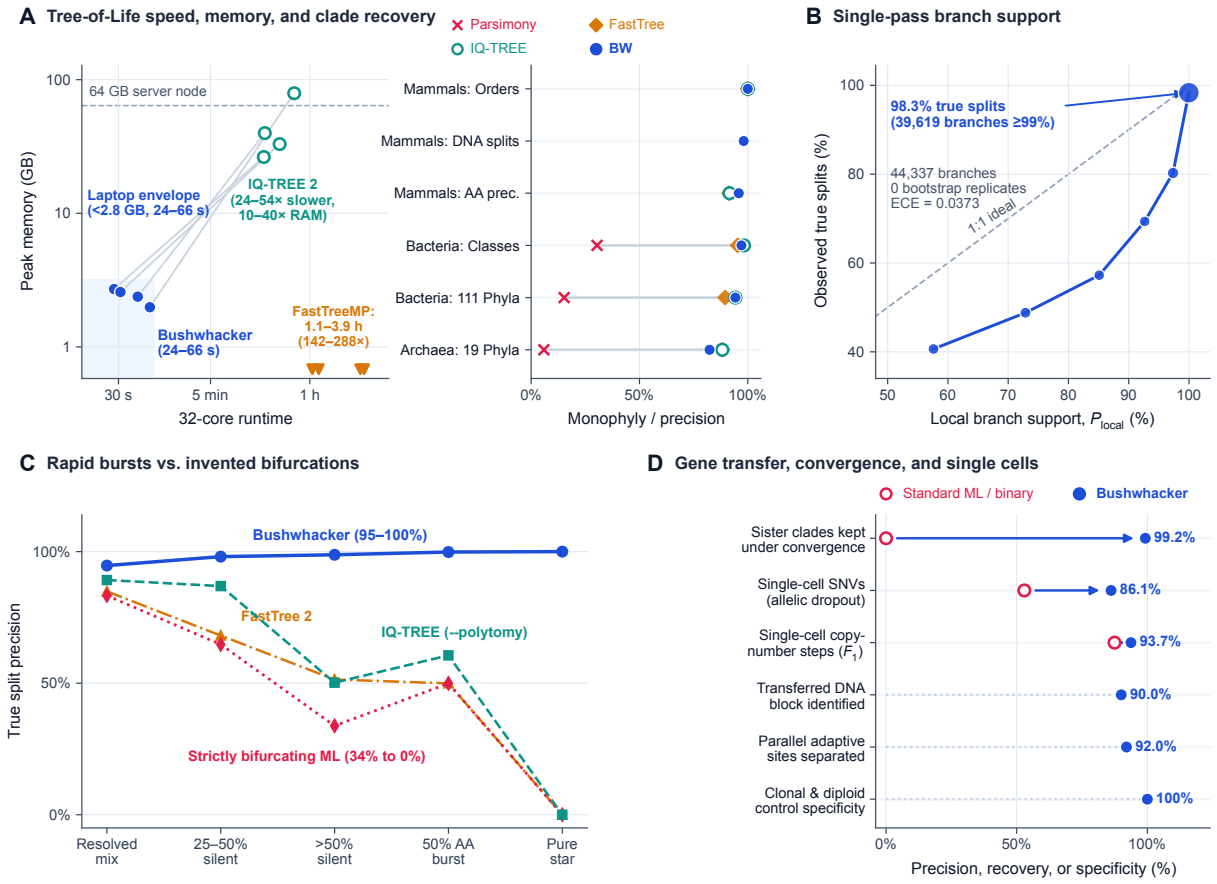


Figure 2: **When trees hold and when bifurcations fail across genomes, networks, and single cells.** (**A**) Thirty-two-core runtime and peak memory across four Tree-of-Life supermatrices (882 mammals, 543, 864 bp DNA and 181, 288 amino acids; 1, 672 archaea; 1, 000 bacteria; Bushwhacker, 24–66 s and 1.98–2.71 GB versus 26–79 GB for IQ-TREE 2; triangles show FastTreeMP runtimes) alongside taxonomic monophyly and split precision against IQ-TREE 2, FastTreeMP, and an unweighted-parsimony ablation (normalized Robinson–Foulds distance within 0.5–2.5 percentage points: nRF = 0.081–0.247 versus 0.074–0.236; table S11). (**B**) Bootstrap-free conditional local branch support (P_{local}) versus observed split accuracy across 44, 337 branches (ECE = 0.0373; 98.3% true at $P_{\text{local}} \geq 99\%$, 40.6%–80.3% across intermediate bins; circle area proportional to branch count; table S3). (**C**) Split precision across 36, 120 simulated nucleotide, protein, and coalescent histories (0.00–0.18 versus 23–61 false splits per rapid burst; table S2). (**D**) Clade retention, event recovery, and control specificity across 800 benchmarks of horizontal transfer, parallel convergence, and error-prone single-cell somatic states (400 clonal and diploid controls; tables S4 and S5).

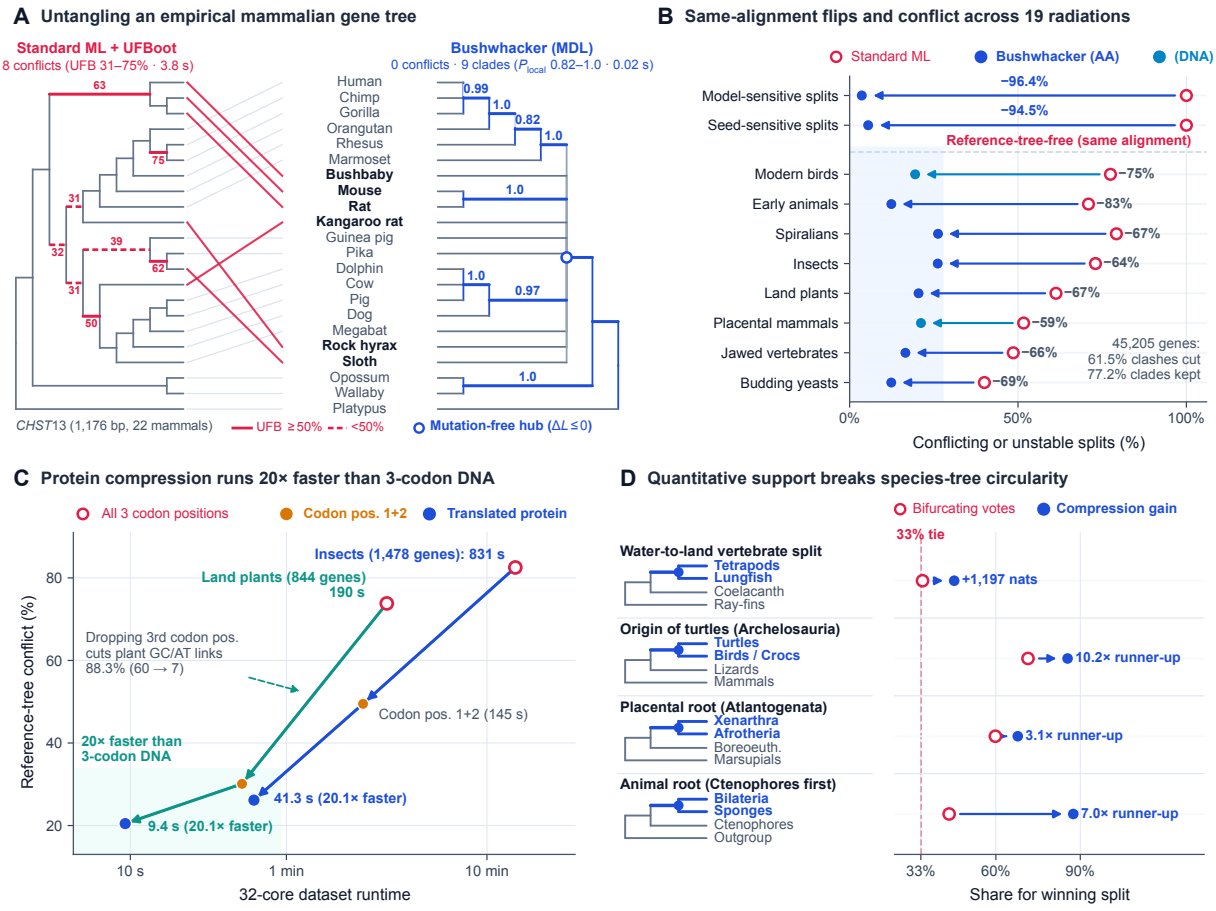


Figure 3: **Mutation-free stems account for widespread gene-tree conflict and model sensitivity.** (**A**) Mammalian *CHST13* tanglegram (1, 176 bp, 22 species) contrasting standard maximum likelihood annotated with ultrafast bootstrap support (8 conflicting splits in crimson: solid ≥ 50%, dashed < 50% UFBoot; 3.8 s on 1 core; fig. S7) against Bushwhacker (0 conflicts; 9 true clades at $P_{\text{local}} = 0.82$–1.0 in 0.02 s; open circle, $\Delta L \leq 0$ placental hub). (**B**) Reference-free controls showing 96.4% removal of model-sensitive splits (7, 200 genes) and 94.5% removal of seed-sensitive splits across 20 RAxML/IQ-TREE runs per locus (83, 180 trees across 4, 159 alignments; top; table S6), above conflict reduction across 45, 205 genes in 19 eukaryotic radiations (61.5% clashes removed; 77.2% shared clades retained; bottom). (**C**) Runtime and conflict across coding DNA, codons 1 + 2, and translated proteins in plants (844 genes) and insects (1, 478 genes), cutting saturated plant links by 88.3% (60 to 7). (**D**) Competing quartet vote shares before (strictly bifurcating counts) and after ($\Delta L > 0$ weighting) breaking species-tree circularity (dashed line, 33% three-way tie).

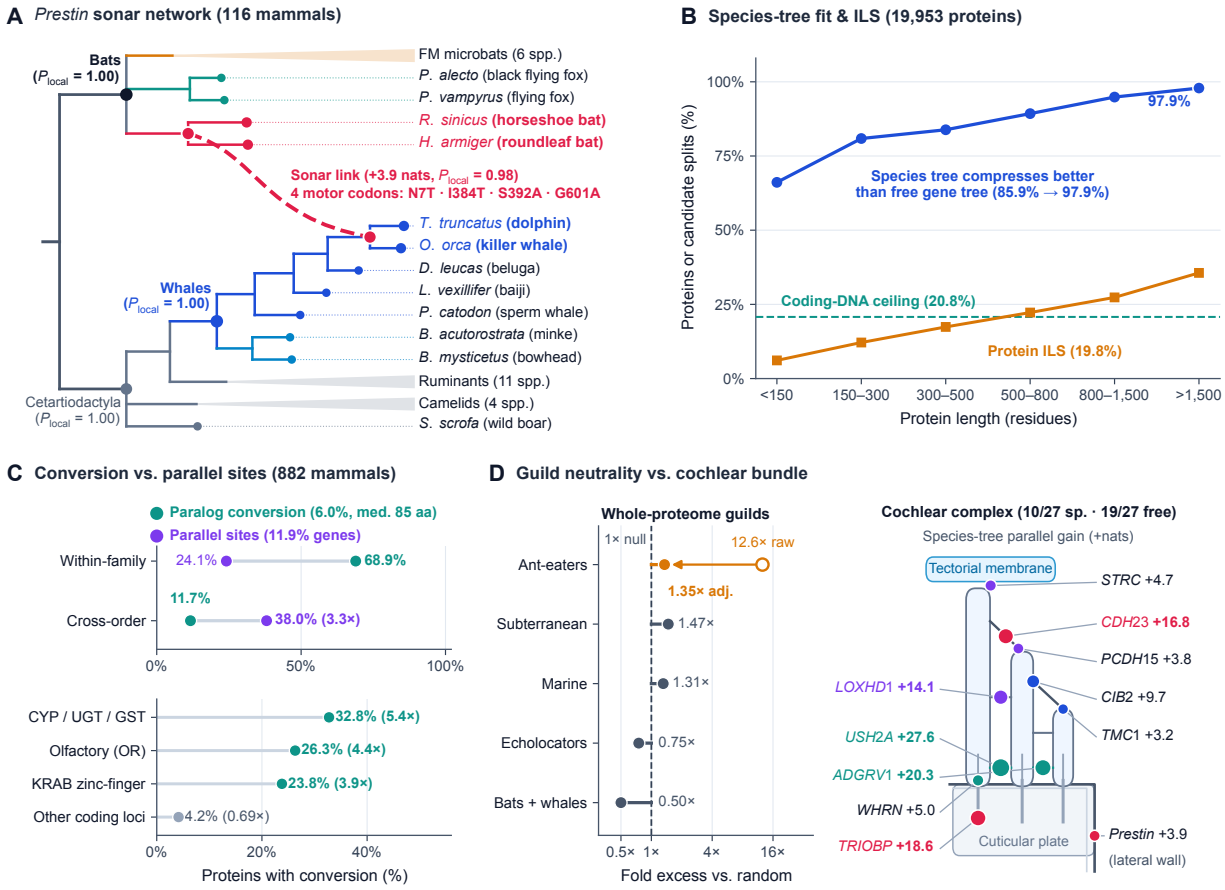


Figure 4: **Unified mammalian census separates organismal inheritance, gene conversion, and convergence.** (**A**) Mammalian *Prestin* (*SLC26A5*, 116 species, 2,235 bp; Laurasiatheria shown) preserves monophyletic bats and Cetartiodactyla ($P_{\text{local}} = 1.00$) while isolating the echolocator convergence link (+3.9 nats, $P_{\text{local}} = 98.0\%$) across four motor-domain codons (N7T, I384T, S392A, G601A). (**B**) Proteins compressing better on the species tree than a free gene tree (85.9% overall; 97.9% above 1,500 residues, blue) and species-tree branch incomplete lineage sorting ($\Delta L > 0$: 19.8% overall, amber; 20.8% coding-DNA ceiling, teal) across 19,953 TOGA2 proteins in 882 mammals. (**C**) Separation of 8,670 species-tree links into contiguous paralog conversion (6.0% of proteins, 2,251 tracts, median 85 residues; 68.9% intra-family, 3.0–6.6× enriched in tandem clusters) and scattered parallel-site links (11.9%, 6,419 links; 38.0% cross-order). (**D**) Guild-wide neutrality (0.50–1.47×; ant-eating 1.35× length-adjusted; left) versus focal convergence across 10/27 species-tree (19/27 free-tree) cochlear proteins (right; table S7).

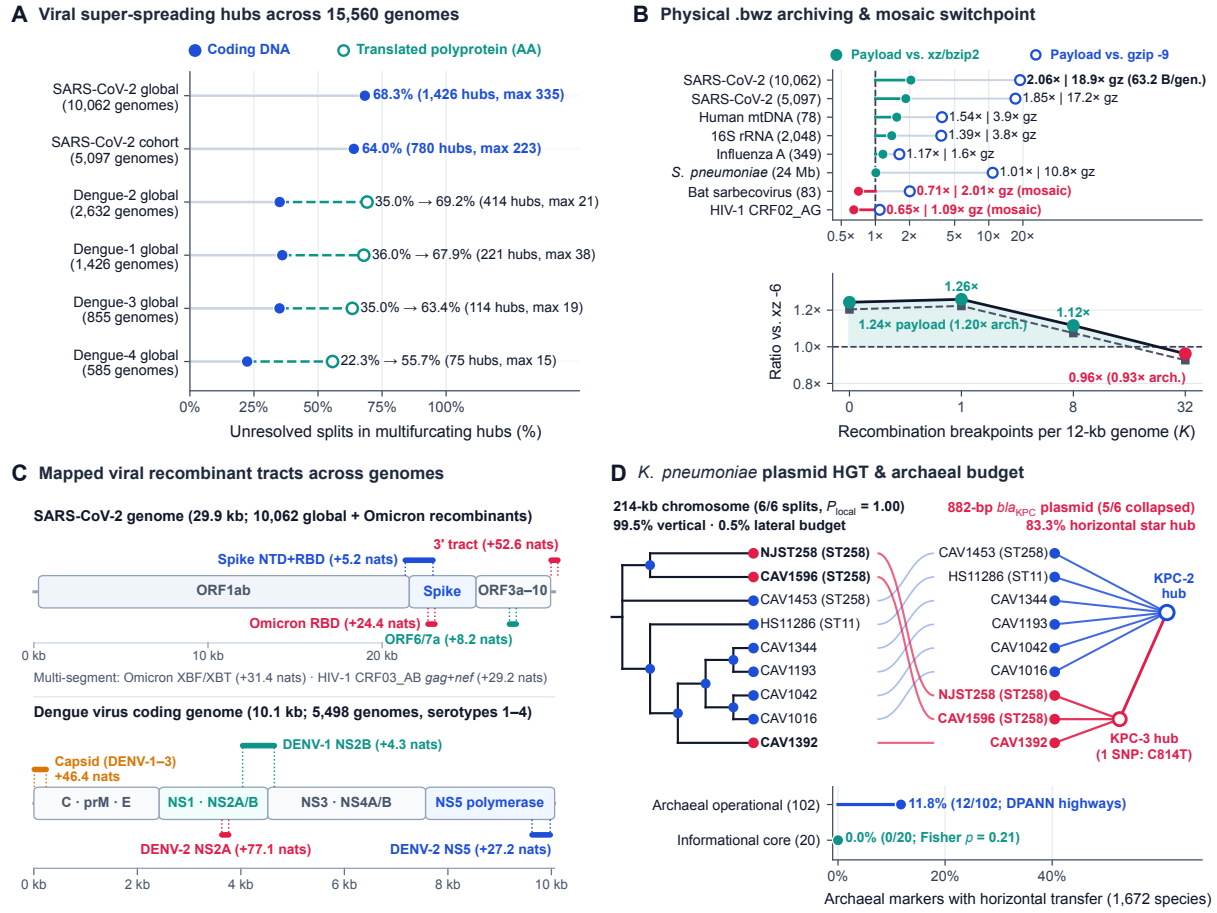


Figure 5: **Viral outbreaks, physical genome archiving, recombination, and prokaryotic gene transfer budgets.** (**A**) Multifurcating outbreak hubs and largest burst sizes across 10, 062 global SARS-CoV-2 (68.3% unresolved; 1, 426 hubs up to 335 descendants) and 5, 498 dengue genomes (serotypes 1–4) in coding DNA versus translated polyproteins. (**B**) Lossless .bwz sequence payload and full archive reductions over xz -6 or bzip2 -9 (up to 2.06× payload) and gzip -9 (up to 18.9×; 38.5 B/genome payload, 63.2 B/genome archive on SARS-CoV-2; top), compared against crossover below 1× under breakpoint titrations ($K = 0 \rightarrow 32$; bottom), bat sarbecoviruses (0.71×), and recombinant HIV-1 (0.65×). (**C**) Mapped mosaic tracts along the 29.9-kb SARS-CoV-2 and 10.1-kb dengue genomes. (**D**) Nine carbapenem-resistant *Klebsiella pneumoniae* isolates across the 214-kb chromosome (6/6 splits, 99.5% vertical budget) versus the 882-bp *blaKPC* plasmid (5/6 splits collapsed into C814T star hubs; top), and archaeal transfer rates across 122 markers in 1, 672 species (11.8% operational versus 0.0% informational, Fisher $p = 0.21$; bottom).

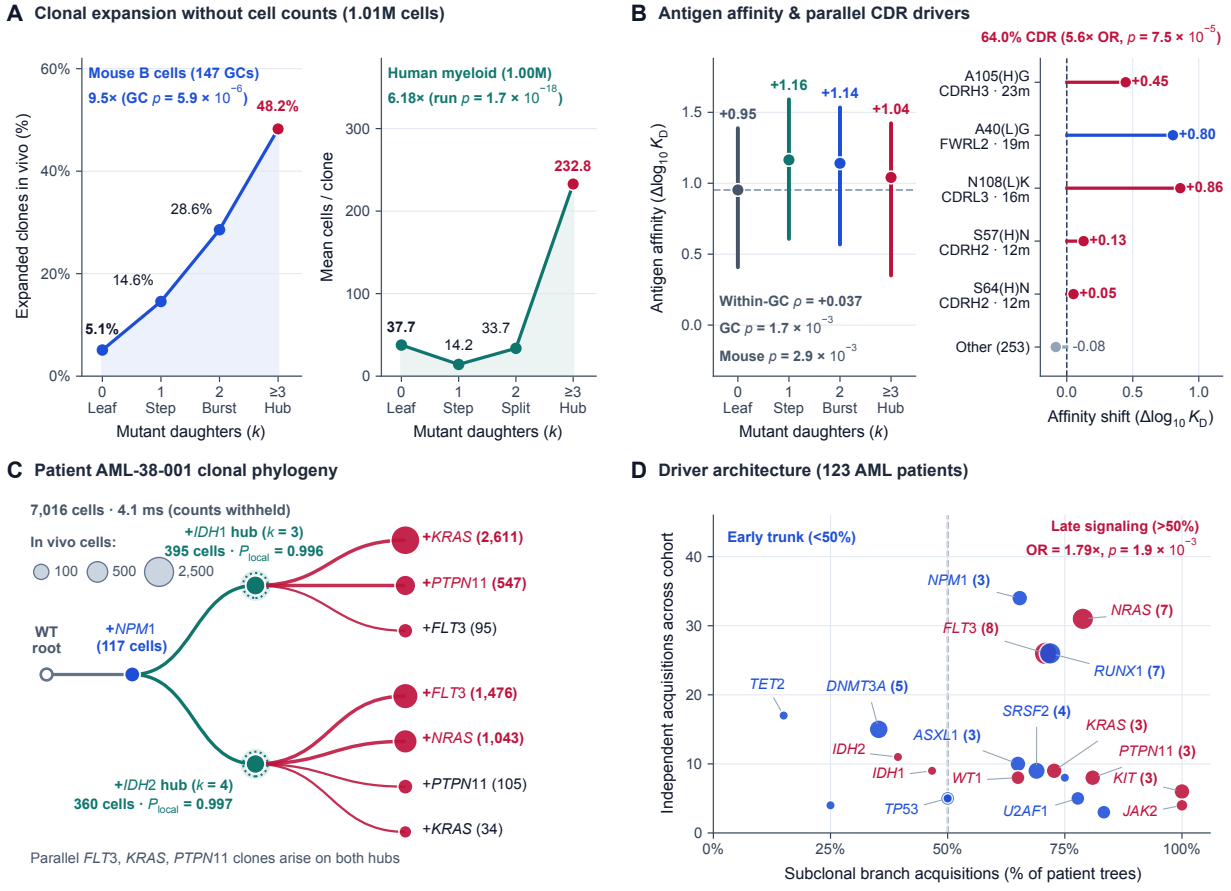


Figure 6: **Somatic compression geometry detects clonal selection across immune and cancer cells.** (**A**) With cell counts withheld, mutant daughter degree k predicts in vivo clonal expansion across 147 mouse germinal centers (10,429 B cells; 5.1% $\rightarrow$ 48.2%, 9.5×, paired Wilcoxon $p = 5.9 \times 10^{-6}$; left) and 168 myeloid single-cell cohorts (1,001,609 cells; 37.7 $\rightarrow$ 232.8 cells/clone, 6.18×, $p = 1.7 \times 10^{-18}$; right). (**B**) Antigen affinity ($\Delta \log_{10} K_D$) across B-cell hubs (within-GC $p = 1.7 \times 10^{-3}$, within-mouse $p = 2.9 \times 10^{-3}$; left) and cross-mouse parallel mutations (64.0% in complementarity-determining regions versus 24.1% background, 5.6× odds ratio, $p = 7.5 \times 10^{-5}$; right). (**C**) Patient AML-38-001 phylogeny (7,016 cells; 4.1 ms without counts; circle area proportional to held-out cell count): an *NPM1* trunk splits into *IDH1* ($P_{\text{local}} = 0.996$, $k = 3$) and *IDH2* ($P_{\text{local}} = 0.997$, $k = 4$) resistance hubs. (**D**) Early epigenetic versus late signaling drivers across 123 AML patients (OR = 1.79, Fisher $p = 1.9 \times 10^{-3}$; counts denote within-patient recurrences).

References and Notes

1. W. M. Fitch, Toward defining the course of evolution: minimum change for a specific tree topology. *Systematic Zoology* **20** (4), 406–416 (1971), doi:10.2307/2412116.

2. J. Felsenstein, Evolutionary trees from DNA sequences: a maximum likelihood approach. *Journal of Molecular Evolution* **17** (6), 368–376 (1981), doi:10.1007/BF01734359.

3. N. Saitou, M. Nei, The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution* **4** (4), 406–425 (1987), doi:10.1093/oxfordjournals.molbev.a040454.

4. M. N. Price, P. S. Dehal, A. P. Arkin, FastTree 2 – approximately maximum-likelihood trees for large alignments. *PLoS ONE* **5** (3), e9490 (2010), doi:10.1371/journal.pone.0009490.

5. A. M. Kozlov, D. Darriba, T. Flouri, B. Morel, A. Stamatakis, RAxML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference. *Bioinformatics* **35** (21), 4453–4455 (2019), doi:10.1093/bioinformatics/btz305.

6. B. Q. Minh, *et al.*, IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution* **37** (5), 1530–1534 (2020), doi:10.1093/molbev/msaa015.

7. J. Rissanen, Modeling by shortest data description. *Automatica* **14** (5), 465–471 (1978), doi:10.1016/0005-1098(78)90005-5.

8. J. Rissanen, A universal prior for integers and estimation by minimum description length. *The Annals of Statistics* **11** (2), 416–431 (1983), doi:10.1214/aos/1176346150.

9. J. Felsenstein, Confidence limits on phylogenies: an approach using the bootstrap. *Evolution* **39** (4), 783–791 (1985), doi:10.1111/j.1558-5646.1985.tb00420.x.

10. S. L. Kosakovsky Pond, D. Posada, M. B. Gravenor, C. H. Woelk, S. D. W. Frost, Automated phylogenetic detection of recombination using a genetic algorithm. *Molecular Biology and Evolution* **23** (10), 1891–1901 (2006), doi:10.1093/molbev/msl051.

11. A. Rokas, S. B. Carroll, Bushes in the tree of life. *PLoS Biology* **4** (11), e352 (2006), doi: 10.1371/journal.pbio.0040352.

12. X.-X. Shen, C. T. Hittinger, A. Rokas, Contentious relationships in phylogenomic studies can be driven by a handful of genes. *Nature Ecology & Evolution* **1** (5), 0126 (2017), doi: 10.1038/s41559-017-0126.

13. S. Abadi, D. Azouri, T. Pupko, I. Mayrose, Model selection may not be a mandatory step for phylogeny reconstruction. *Nature Communications* **10** (1), 934 (2019), doi: 10.1038/s41467-019-08822-w.

14. J. Felsenstein, Cases in which parsimony or compatibility methods will be positively misleading. *Systematic Zoology* **27** (4), 401–410 (1978), doi:10.2307/2412923.

15. Z. Yang, Maximum likelihood phylogenetic estimation from DNA sequences with variable rates over sites: approximate methods. *Journal of Molecular Evolution* **39** (3), 306–314 (1994), doi:10.1007/BF00160154.

16. S. Q. Le, O. Gascuel, An improved general amino acid replacement matrix. *Molecular Biology and Evolution* **25** (7), 1307–1320 (2008), doi:10.1093/molbev/msn067.

17. W. F. Doolittle, Phylogenetic classification and the universal tree. *Science* **284** (5423), 2124–2128 (1999), doi:10.1126/science.284.5423.2124.

18. M. Worobey, A. Rambaut, E. C. Holmes, Widespread intra-serotype recombination in natural populations of dengue virus. *Proceedings of the National Academy of Sciences* **96** (13), 7352–7357 (1999), doi:10.1073/pnas.96.13.7352.

19. Y. Liu, *et al.*, Convergent sequence evolution between echolocating bats and dolphins. *Current Biology* **20** (2), R53–R54 (2010), doi:10.1016/j.cub.2009.11.058.

20. J. Parker, *et al.*, Genome-wide signatures of convergent evolution in echolocating mammals. *Nature* **502** (7470), 228–231 (2013), doi:10.1038/nature12511.

21. Z. Zou, J. Zhang, No genome-wide protein sequence convergence for echolocation. *Molecular Biology and Evolution* **32** (5), 1237–1241 (2015), doi:10.1093/molbev/msv014.

22. Y. Turakhia, *et al.*, Pandemic-scale phylogenomics reveals the SARS-CoV-2 recombination landscape. *Nature* **609** (7929), 994–997 (2022), doi:10.1038/s41586-022-05189-9.

23. Y. V. Malovichko, *et al.*, Accurate, comprehensive gene annotation and ortholog identification across thousands of vertebrate genomes with TOGA2. *bioRxiv* (2026), doi:10.64898/2026.06.30.735536.

24. R. Jain, M. C. Rivera, J. A. Lake, Horizontal gene transfer among genomes: the complexity hypothesis. *Proceedings of the National Academy of Sciences* **96** (7), 3801–3806 (1999), doi:10.1073/pnas.96.7.3801.

25. D. H. Parks, *et al.*, A complete domain-to-species taxonomy for Bacteria and Archaea. *Nature Biotechnology* **38** (9), 1079–1086 (2020), doi:10.1038/s41587-020-0501-8.

26. W. S. DeWitt, *et al.*, Replaying germinal center evolution on a quantified affinity landscape. *Cell* (2026).

27. K. Morita, *et al.*, Clonal evolution of acute myeloid leukemia revealed by high-throughput single-cell genomics. *Nature Communications* **11** (1), 5327 (2020), doi:10.1038/s41467-020-19119-8.

28. Materials and methods are available as supplementary materials.

29. C. Zhang, R. Nielsen, S. Mirarab, ASTER: A package for large-scale phylogenomic reconstructions. *Molecular Biology and Evolution* **42** (8), msaf172 (2025), doi:10.1093/molbev/msaf172.

30. E. D. Jarvis, *et al.*, Whole-genome analyses resolve early branches in the tree of life of modern birds. *Science* **346** (6215), 1320–1331 (2014), doi:10.1126/science.1253451.

31. R. O. Prum, *et al.*, A comprehensive phylogeny of birds (Aves) using targeted next-generation DNA sequencing. *Nature* **526** (7574), 569–573 (2015), doi:10.1038/nature15697.

32. N. J. Wickett, *et al.*, Phylotranscriptomic analysis of the origin and early diversification of land plants. *Proceedings of the National Academy of Sciences* **111** (45), E4859–E4868 (2014), doi:10.1073/pnas.1323926111.

33. B. Misof, *et al.*, Phylogenomics resolves the timing and pattern of insect evolution. *Science* **346** (6210), 763–767 (2014), doi:10.1126/science.1257570.

34. M.-Y. Chen, D. Liang, P. Zhang, Selecting question-specific genes to reduce incongruence in phylogenomics: a case study of jawed vertebrate backbone phylogeny. *Systematic Biology* **64** (6), 1104–1120 (2015), doi:10.1093/sysbio/syv059.

35. M. L. Borowiec, E. K. Lee, J. C. Chiu, D. C. Plachetzki, Extracting phylogenetic signal and accounting for bias in whole-genome data sets supports the Ctenophora as sister to remaining Metazoa. *BMC Genomics* **16**, 987 (2015), doi:10.1186/s12864-015-2146-4.

36. N. V. Whelan, *et al.*, Ctenophore relationships and their placement as the sister group to all other animals. *Nature Ecology & Evolution* **1** (11), 1737–1746 (2017), doi:10.1038/s41559-017-0331-3.

37. S. Song, L. Liu, S. V. Edwards, S. Wu, Resolving conflict in eutherian mammal phylogeny using phylogenomics and the multispecies coalescent model. *Proceedings of the National Academy of Sciences* **109** (37), 14942–14947 (2012), doi:10.1073/pnas.1211733109.

38. A. J. Shultz, T. B. Sackton, Immune genes are hotspots of shared positive selection across birds and mammals. *eLife* **8**, e41815 (2019), doi:10.7554/eLife.41815.

39. G. W. C. Thomas, M. W. Hahn, Determining the null model for detecting adaptive convergence from genomic data: a case study using echolocating mammals. *Molecular Biology and Evolution* **32** (5), 1232–1236 (2015), doi:10.1093/molbev/msv013.

40. P.-A. Christin, *et al.*, Adaptive evolution of C_4 photosynthesis through recurrent lateral gene transfer. *Current Biology* **22** (5), 445–449 (2012), doi:10.1016/j.cub.2012.01.054.

41. S. Lytras, *et al.*, Exploring the natural origins of SARS-CoV-2 in the light of recombination. *Genome Biology and Evolution* **14** (2), evac018 (2022), doi:10.1093/gbe/evac018.

42. K. L. Wyres, M. M. C. Lam, K. E. Holt, Population genomics of *Klebsiella pneumoniae*. *Nature Reviews Microbiology* **18** (6), 344–359 (2020), doi:10.1038/s41579-019-0315-1.

43. C. J. Castelle, J. F. Banfield, Major new microbial groups expand diversity and alter our understanding of the tree of life. *Cell* **172** (6), 1181–1197 (2018), doi:10.1016/j.cell.2018.02.016.

44. D. McDonald, *et al.*, An improved Greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of bacteria and archaea. *The ISME Journal* **6** (3), 610–618 (2012), doi:10.1038/ismej.2011.139.

45. J. M. J. Tas, *et al.*, Visualizing antibody affinity maturation in germinal centers. *Science* **351** (6277), 1048–1054 (2016), doi:10.1126/science.aad3439.

46. W. S. DeWitt, L. Mesin, G. D. Victora, V. N. Minin, F. A. Matsen, IV, Using genotype abundance to improve phylogenetic inference. *Molecular Biology and Evolution* **35** (5), 1253–1265 (2018), doi:10.1093/molbev/msy020.

47. E. Sollier, J. Kuipers, K. Takahashi, N. Beerenwinkel, K. Jahn, COMPASS: joint copy number and mutation phylogeny reconstruction from amplicon single-cell sequencing data. *Nature Communications* **14** (1), 4921 (2023), doi:10.1038/s41467-023-40378-8.

48. S. Salehi, *et al.*, Cancer phylogenetic tree inference at scale from 1000s of single cell genomes. *Peer Community Journal* **3**, e63 (2023), doi:10.24072/pcjournal.292.

49. S. L. Kosakovsky Pond, *et al.*, Bushwhacker: Evolutionary history from genomes to single cells by lossless compression, version 1.0, GitHub (2026), `https://github.com/veg/bushwhacker`.

Acknowledgments

We thank collaborators and open-data consortia for making multi-scale genomic and single-cell cohorts publicly available.

Funding: Supported by grants from the National Institutes of Health and the National Science Foundation (S.L.K.P. and A.N.).

Author contributions: Conceptualization: S.L.K.P., D.P.M., A.N. Methodology and software: S.L.K.P., D.P.M., S.J.K.P., S.W. Investigation, validation, and visualization: S.L.K.P., D.P.M., S.J.K.P., S.W., D.C., A.N. Writing—original draft and review & editing: S.L.K.P., D.P.M., S.J.K.P., S.W., D.C., A.N.

Competing interests: There are no competing interests to declare.

Data, code and materials availability: The open-source Rust and WebAssembly implementation of Bushwhacker (*49*), the .bwz lossless archival codec, interactive browser application (`https://veg.github.io/bushwhacker/`), reproducibility scripts, and empirical and ground-truth validation tables spanning more than 81,000 empirical alignments, 1.01 million single cells, and 37,020 simulated evolutionary histories are archived at `https://github.com/veg/bushwhacker` and bundled in the Zenodo v1.0 replication archive (`bushwhacker_science_zenodo_v1.0`; see (*28*) for accessions and DOIs). No physical materials were generated in this work.

Supplementary materials

Materials and Methods

Supplementary Text

Figs. S1 to S7

Tables S1 to S11

References *(50–91)*